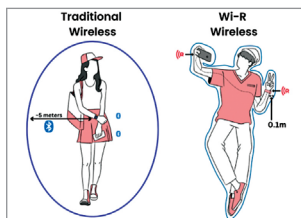




wireless signal that has a range of 2-5 metres in a conductor while less than 10cm around the conductor. This technology utilises a human body as a conductor and confines the signal around



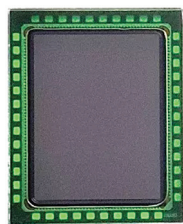
the user's body. It can allow engineers to design devices that provide robustness, efficiency, and touch selectivity of the signal. Moreover, it enhances the safety of the system by having a short transmission range from the body and featuring mathematical security that comes from encryption.

Ixana

<https://ixana.ai>

Low-light image sensor

Compact Mira050 by ams OSRAM is a 0.5-megapixel CMOS image sensor suitable for wearable and mobile devices. It offers high sensitivity to light in the visible and NIR spectrum, thus requiring less illumination to help extend runtime of a battery powered device. The



CMOS sensor can enable engineers to build compact wearable devices, and offer power savings at high-quantum efficiency for smart glasses, VR headsets, etc. The Mira050 is suitable for applications such as eye tracking, gesture tracking, and contextual awareness in AR/VR/MR headsets, object detection in robots, and 3D depth sensing for face recognition in smart door locks.

ams OSRAM

<https://ams-osram.com>

Energy harvesting standalone beacon

BlueRange has released a new beacon with energy harvesting capability. The new beacons combine photovoltaics and Bluetooth beacons to offer a battery-free, standalone system. The

beacon is capable of harvesting a light irradiation of 200 lux to power a device



and send its sensor data. The beacon can be integrated with sensors to sense various parameters such as carbon-dioxide,

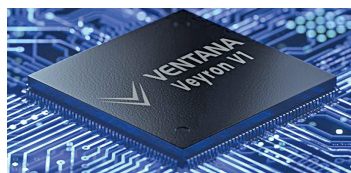
temperature, air pressure, humidity, etc and transmit data to a controller or node. The BlueRange beacons automatically form a mesh connection as soon as they are defined as the same network. The mesh is self-organising and self-healing through the best possible connections between devices.

Bluerange

<https://www.bluerange.io>

High-performance RISC-V processor

Ventana's Veyron V1 is the highest-performance RISC-V processor for data centres running at 3.6GHz in 5nm. It is also the first RISC-V processor to provide single thread performance that is competitive with the latest incumbent processors for data centres, automotive, 5G, AI, and client applications. The



Veyron V1's efficient microarchitecture enables the highest single socket performance, which when combined with RISC-V's open and extensible architecture enables customer innovation and workload optimisation. This results in further workload efficiency gains through domain-specific acceleration that extends Moore's Law to deal with the emerging energy and thermal constraints for data centres.

Ventana

<https://www.ventanamicro.com>

Low-power, compact, accurate GNSS IC

Synaptics SYN4778 is one of the lowest power, compact, and highly

accurate global navigation satellite system (GNSS) IC for IoT devices. The SYN4778 offers up to 80% reduction in power consumption while having up to 50% higher accuracy in a 30% smaller package when compared to most other GNSS ICs available today. The small



size of GNSS IC enables designers to employ highly accurate location based services (LBS) for smaller IoT devices such as wearables and asset trackers. When compared to most existing solutions the SYN4778 will enable designers to build LBS equipment with higher battery life, while also having a smaller product size and enhanced performance. It is suitable for IoT devices such as wearables as well as mobile accessories, asset tagging and tracking systems, drones, transportation navigation, etc.

Synaptics

<https://www.synaptics.com>

Silicon photonics optical gyroscope

ANELLO Photonics SIPHOG is the world's first silicon photonics optical gyroscope. The SIPHOG is a low-noise and low-drift optical gyroscope smart sensor leveraging a photonic



IC technology.

This makes it the smallest optical gyroscope in the world, which enables reliable and accurate navigation

and positioning across many autonomous applications. The SIPHOG can enable companies to design stable, accurate, and robust sensors for various autonomous applications. It features Anello's Sensor Fusion Engine, which delivers a full end-to-end offering to customers in various fields of autonomy where position, navigation, and timing are essential.

ANELLO Photonics

<https://www.anellophotonics.com>